

One-dimensional Turbulence Model (notes)

Sienna White

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Governing equations for flow

We first discuss the derivation of the equations used for our one-dimensional water column model.

Reynolds-averaged Navier Stokes

We start with the Reynolds-Averaged Navier Stokes, which reads:

$$\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = \frac{\partial}{\partial x_k} \overline{u_k u_i} - \frac{1}{\rho_0} \frac{\partial \bar{p}}{\partial x_i} - g \delta_{i3} + \nu \frac{\partial^2}{\partial x_j \partial x_j} \bar{u}_i \quad (1)$$

where flow is considered to be incompressible (solenoidal),

$$\frac{\partial u_i}{\partial x_i} = 0. \quad (2)$$

This equation in the $\hat{e}_1$ direction, expanded, reads

$$\frac{\partial \bar{u}}{\partial t} + \bar{u} \frac{\partial \bar{u}}{\partial x} + \bar{v} \frac{\partial \bar{u}}{\partial y} + \bar{w} \frac{\partial \bar{u}}{\partial z} = \frac{\partial}{\partial x} \overline{u' u'} + \frac{\partial}{\partial y} \overline{v' u'} + \frac{\partial}{\partial z} \overline{w' u'} - \frac{1}{\rho_0} \frac{\partial \bar{p}}{\partial x} + \nu \left(\frac{\partial^2 \bar{u}}{\partial x^2} + \frac{\partial^2 \bar{u}}{\partial y^2} + \frac{\partial^2 \bar{u}}{\partial z^2} \right).$$

We neglect advection and gradients in the x and y directions and take $w = 0$ (established by the boundary layer approximation),

$$\frac{\partial \bar{u}}{\partial t} + \cancel{\bar{u} \frac{\partial \bar{u}}{\partial x}} + \cancel{\bar{v} \frac{\partial \bar{u}}{\partial y}} + \cancel{\bar{w} \frac{\partial \bar{u}}{\partial z}} = \cancel{\frac{\partial}{\partial x} \overline{u' u'}} + \cancel{\frac{\partial}{\partial y} \overline{v' u'}} + \frac{\partial}{\partial z} \overline{w' u'} - \frac{1}{\rho_0} \frac{\partial \bar{p}}{\partial x} + \nu \left(\cancel{\frac{\partial^2 \bar{u}}{\partial x^2}} + \cancel{\frac{\partial^2 \bar{u}}{\partial y^2}} + \frac{\partial^2 \bar{u}}{\partial z^2} \right)$$

We now have a simplified equation for Reynolds-averaged flow in the $\hat{e}_1$ direction.

$$\frac{\partial \bar{u}}{\partial t} = \frac{\partial}{\partial z} \overline{w' u'} - \frac{1}{\rho_0} \frac{\partial \bar{p}}{\partial x} + \nu \frac{\partial^2 \bar{u}}{\partial z^2}. \quad (3)$$

Next, we apply the Boussinesq assumption, which allows us to re-write the Reynolds Stress as a function of the mean velocity gradient. Note that the k term drops out as our Reynolds stress is off-diagonal,

$$\overline{w' u'} = \mu_t \left(\frac{\partial \bar{u}}{\partial z} + \frac{\partial \bar{w}}{\partial x} \right) - \cancel{\frac{2}{3} \rho_0 \bar{\epsilon}}_{13} = \mu_t \left(\frac{\partial \bar{u}}{\partial z} + \cancel{\frac{\partial \bar{w}}{\partial x}} \right) = \mu_t \frac{\partial \bar{u}}{\partial z}$$

We plug our expression for the Reynold's stress into Eq. 3,

$$\frac{\partial \bar{u}}{\partial t} = \frac{\partial}{\partial z} \left(\mu_t \frac{\partial \bar{u}}{\partial z} \right) - \frac{1}{\rho_0} \frac{\partial \bar{p}}{\partial x} + \nu \frac{\partial^2 \bar{u}}{\partial z^2}$$

$$\frac{\partial \bar{u}}{\partial t} = \mu_t \frac{\partial^2 \bar{u}}{\partial z^2} - \frac{1}{\rho_0} \frac{\partial \bar{p}}{\partial x} + \nu \frac{\partial^2 \bar{u}}{\partial z^2}$$

We can group the turbulent and molecular viscosity terms,

$$\frac{\partial \bar{u}}{\partial t} = -\frac{1}{\rho_0} \frac{\partial \bar{p}}{\partial x} + (\nu_t + \nu) \frac{\partial^2 \bar{u}}{\partial z^2}$$

however, since we assume that turbulent viscosity is far greater than molecular viscosity,

$$\nu_t \gg \nu$$

we often neglect molecular viscosity, leaving us with

$$\frac{\partial \bar{u}}{\partial t} = -\frac{1}{\rho_0} \frac{\partial \bar{p}}{\partial x} + \nu_t \frac{\partial^2 \bar{u}}{\partial z^2}.$$

After applying the Boussinesq assumption, we've escaped the need to resolve the Reynold's Stress, however, in discretizing this equation, we still need to approximate this turbulent viscosity ν_t . Hence the need for a closure scheme.

Turbulence closure

Similarity theory tells us that ν_t , which has units of [length² time⁻¹] can be scaled by a length scale times a velocity scale,

$$\nu_t \approx \ell_T U_T$$

Here I am using the subscript T to denote these quantities being *turbulent scales*. We estimate the velocity scale using the square root of the total turbulent kinetic energy,

$$\text{TKE} = \text{tr}(\overline{u_i u_j}) = (\overline{u'^2} + \overline{v'^2} + \overline{w'^2})$$

$$q = \sqrt{\overline{u'^2} + \overline{v'^2} + \overline{w'^2}}$$

$$\nu_t \approx \ell_T q$$

Again, ν_t is a function $\nu_t(z, t)$ that dissipates energy from the mean velocity (due to subscale turbulence).

The TKE equation

We can derive an equation for the turbulent kinetic energy, q^2 . This process is fairly intensive so I will put it in an appendix or something. The equation for the turbulent kinetic energy q^2 given by,

$$\underbrace{\frac{\partial q^2}{\partial t}}_1 = \underbrace{\frac{\partial}{\partial z} \left(k_q \frac{\partial q^2}{\partial z} \right)}_2 + \underbrace{2P}_2 + \underbrace{2B}_3 - \underbrace{2\epsilon}_4$$

Moving from left to right,

$$[1] \quad \frac{\partial q^2}{\partial t}$$

is **unsteadiness**, or the change in turbulent kinetic energy over time;

$$[2] \quad \frac{\partial}{\partial z} \left(k_q \frac{\partial q^2}{\partial z} \right)$$

is the **turbulent diffusion**, or a measure of how turbulent energy “diffuses” throughout the fluid flow. This neither creates nor destroys turbulence— just moves it. Next is

$$[3] \quad P = \overline{u_i u_k} \frac{\partial u_i}{\partial x_k},$$

production, also called **shear production**. We expand this for our simplified scenario (neglecting gradients of x , y , $w = 0$)

$$P = \overline{u'w'} \frac{\partial u}{\partial z} + \overline{v'w'} \frac{\partial v}{\partial z} + \overline{w'w'} \frac{\partial w}{\partial z}$$

$$P = \mu_t \left(\frac{\partial \bar{u}}{\partial z} + \frac{\partial \bar{w}}{\partial z} \right) \frac{\partial u}{\partial z} + \mu_t \left(\frac{\partial \bar{v}}{\partial z} + \frac{\partial \bar{w}}{\partial y} \right) \frac{\partial v}{\partial z}$$

$$P = \mu_t \left(\frac{\partial \bar{u}}{\partial z} \right)^2 + \mu_t \left(\frac{\partial \bar{v}}{\partial z} \right)^2$$

Dissipation is a disaster

$$[4] \quad \epsilon = -\nu \left(\frac{\partial u_i}{\partial x_k} \frac{\partial u_i}{\partial x_k} \right)$$

$$\epsilon = -\nu \left(\left(\frac{\partial u}{\partial z} \right)^2 + \left(\frac{\partial v}{\partial z} \right)^2 + \left(\frac{\partial w}{\partial z} \right)^2 \right)$$

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z} \left(k_q \frac{\partial q^2}{\partial z} \right) + \nu_t \frac{\partial^2 u^2}{\partial z^2} + 2 \frac{g}{\rho_0} \kappa_p \frac{\partial p}{\partial z} - 2 \frac{(q^2)^{3/2}}{B_1 l}$$

Similar to our simplifications before, we can comfortably neglect advection of TKE as long as the time scale of dissipation is shorter than the advection time scale T_a over one grid cell in the horizontal.

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z} \left(k_q \frac{\partial q^2}{\partial z} \right) - \overline{u_i u_k} \frac{\partial u_i}{\partial x_k} + 2B - \nu \overline{\frac{\partial u_i}{\partial x_k} \frac{\partial u_i}{\partial x_k}}$$

Boundary layer approximation for environmental flow

The equations used in our model assume that in our environment, the lateral length scales L_x and L_y are far greater than the vertical length scale (or depth) L_z . This means that gradients in the x and y directions can be seen as very small, or negligible. We maintain gradients in the pressure direction as these drive the flow and are assumed not to be functions of x .

$$\begin{aligned} \frac{\partial u}{\partial t} + u \cancel{\frac{\partial u}{\partial x}} + v \cancel{\frac{\partial u}{\partial y}} + w \frac{\partial u}{\partial z} &= -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu_t \left(\cancel{\frac{\partial^2 u}{\partial x^2}} + \cancel{\frac{\partial^2 u}{\partial y^2}} + \frac{\partial^2 u}{\partial z^2} \right) \\ \frac{\partial u}{\partial t} + w \frac{\partial u}{\partial z} &= -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu_t \left(\frac{\partial^2 u}{\partial z^2} \right) \end{aligned}$$

We also can prove that $w = 0$, simplifying our equation to

$$\frac{\partial u}{\partial t} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu_t \left(\frac{\partial^2 u}{\partial z^2} \right) \quad (4)$$

Boussinesq gradient diffusion hypothesis

Boussinesq attempted to simplify the mess of the Reynold's stress tensor $\overline{u_i u_j}$ by reasoning that these correlation terms would act like a stress tensor in a flow.

In deriving the Navier-Stokes we define a stress tensor as

$$-(\tau_{ij} + P\delta_{ij}) = -2\mu S_{ij} = -\mu \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right)$$

where τ_{ij} represents shear stresses and $P\delta_{ij}$, or pressure, acts as a body force along the diagonal of the tensor.

If the Reynold's stresses are like shear stresses imposed on flow, we can substitute them in here

$$-(\overline{u_i u_j} + \frac{2}{3} k \delta_{ij}) = -\mu_t \left(\frac{\partial \overline{u_i}}{\partial x_j} + \frac{\partial \overline{u_j}}{\partial x_i} \right)$$

only this turbulent stress tensor would be approximated by some turbulent viscosity μ_t and the

$$\overline{u_i u_j} = \mu_t \left(\frac{\partial \overline{u_i}}{\partial x_j} + \frac{\partial \overline{u_j}}{\partial x_i} \right) - \frac{2}{3} k \delta_{ij}$$

Note that the trace of the RHS would be

$$\mu_t \left(2 \frac{\partial u_i}{\partial x_i} - \frac{2}{3} k \right) = \mu_T \left(-3 \frac{2}{3} k \right) = -2 \mu_T k = -2 \mu_T \left(\frac{1}{2} \overline{u'_i u'_i} \right)$$

The main thing to note here is that using this approximation, we can now represent the Reynold's stress as being driven by gradients in the mean flow. This will help us from a closure perspective.

The two-equation model

Our equation for the turbulent kinetic energy q is now given by,

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z} \left(k_q \frac{\partial q^2}{\partial z} \right) + 2P + 2B - 2\epsilon$$

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z} \left(k_q \frac{\partial q^2}{\partial z} \right) - \overline{u_i u_k} \frac{\partial u_i}{\partial x_k} + 2B - \nu \frac{\partial u_i}{\partial x_k} \frac{\partial u_i}{\partial x_k}$$

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z} \left(k_q \frac{\partial q^2}{\partial z} \right) + \nu_t \frac{\partial u^2}{\partial z} + 2 \frac{g}{\rho_0} \kappa_p \frac{\partial p}{\partial z} - 2 \frac{(q^2)^{3/2}}{B_1 l}$$

We can neglect advection as long as the time scale of dissipation is short than the advection time scale T_a over one grid cell in the horizontal.

The Mellor-Yamada 2.5 closure

The chosen closure scheme for our 1-D turbulent model is the Mellor-Yamada 2.5 scheme. This scheme approximates the eddy viscosity as

$$\nu_t = S_m q L \tag{5}$$

and the turbulent diffusivity as

$$\kappa_q = S_h Q L, \tag{6}$$

where S_m and S_h are empirically-derived stability functions, q is the turbulent kinetic energy, and l is our turbulent length scale.

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z} \left(qlS_q \frac{\partial q^2}{\partial z} \right) + 2 \left(P_s + P_b - \frac{q^3}{B_1 l} \right) \quad (7)$$

$$\frac{\partial q^2 l}{\partial t} = \frac{\partial}{\partial z} \left(qlS_q \frac{\partial q^2 l}{\partial z} \right) + lE_1(P_s + P_b) - \frac{q^3}{B_1 l} W(z) \quad (8)$$

where P_s denotes shear production, and is given by

$$P_s = \nu_T \left(\left(\frac{\partial U}{\partial z} \right)^2 + \left(\frac{\partial V}{\partial z} \right)^2 \right),$$

the buoyancy production, P_b , is calculated as

$$P_b = -\kappa_t \frac{g}{\rho_0} \frac{\partial \bar{\rho}}{\partial z},$$

and finally the turbulent dissipation is given by

$$\epsilon = \frac{q^3}{B_1 l}.$$

The function $W(z)$ is a wall-proximity function that adjusts dissipation as z approaches a boundary. S_q and E_1 are both empirical constants.

Discretization

Advection-Diffusion Equation for Algae

Our governing equation for algae is given by,

$$\frac{\partial C}{\partial t} + w_s \frac{\partial C}{\partial z} = \frac{\partial}{\partial z} \left(\kappa_t \frac{\partial C}{\partial z} \right) + \gamma C \quad (9)$$

where C is the concentration of algae (scalar), γ is some source/loss term, and w_s is the sinking or swimming speed of the algae: w_s can be positive (*Microcystis*) or negative (phytoplankton).

We break down the discretization of Equation 9 by the left hand (time derivative + advection) and right hand (diffusion) sides. The right-hand side is trickiest.

Discretizing diffusion

We first evaluate the diffusion term in space (neglecting γ for now). Note here that diffusion is given by κ_t , which represents turbulent diffusion (a quantity we will solve for in the turbulence closure). Because the subscripts get messy, I'll drop the subscript t , but all diffusion in this equation refers to κ_t .

We evaluate $\kappa_t \frac{\partial C}{\partial z}$ at $i + 1/2, i - 1/2$,

$$\frac{\partial}{\partial z} \left(\kappa_t \frac{\partial C}{\partial z} \right) \rightarrow \frac{1}{\Delta z} \left(\left(\kappa_t \frac{\partial C}{\partial z} \right)_{i+1/2} - \left(\kappa_t \frac{\partial C}{\partial z} \right)_{i-1/2} \right)$$

but evaluate C at $i + 1, i - 1$ (this is where I will now drop the subscript on κ_t),

$$= \frac{1}{\Delta z} \left(\kappa_{i+1/2} \frac{\partial C}{\partial z}_{i+1} - \kappa_{i-1/2} \frac{\partial C}{\partial z}_{i-1} \right).$$

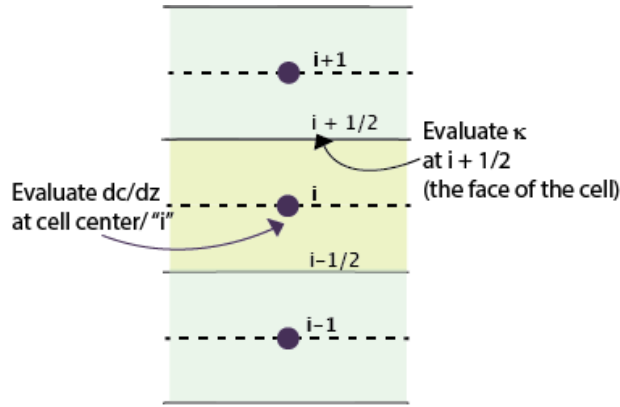


Figure 1: Stencil for κ_t & C

We represent these 1/2 stencil points for κ_t as the average of the two neighboring cells,

$$\kappa_{i+1/2} \approx \frac{\kappa_{i+1} + \kappa_i}{2}$$

$$\kappa_{i-1/2} \approx \frac{\kappa_{i-1} + \kappa_i}{2}$$

And discretize the gradient of C ,

$$\left(\frac{\partial C}{\partial z} \right)_{i+1/2} \approx \frac{C_{i+1} - C_i}{\Delta z}$$

$$\left(\frac{\partial C}{\partial z} \right)_{i-1/2} \approx \frac{C_i - C_{i-1}}{\Delta z}$$

Putting these together,

$$\begin{aligned} \frac{1}{\Delta z} \left(\left(\kappa \frac{\partial C}{\partial z} \right)_{i+1/2} - \left(\kappa \frac{\partial C}{\partial z} \right)_{i-1/2} \right) &\rightarrow \frac{1}{\Delta z} \left(\frac{\kappa_{i+1} + \kappa_i}{2} \frac{C_{i+1} - C_i}{\Delta z} - \frac{\kappa_{t(i-1)} + \kappa_i}{2} \frac{C_i - C_{i-1}}{\Delta z} \right) \\ &= \frac{1}{2\Delta z^2} ((\kappa_{i+1} + \kappa_i)(C_{i+1} - C_i) - (\kappa_{i-1} + \kappa_i)(C_i - C_{i-1})) \end{aligned}$$

Factor according to C ,

$$\begin{aligned} &= C_{i+1} \left(\frac{1}{2\Delta z^2} (\kappa_{i+1} + \kappa_i) \right) \\ &+ C_i \left(\frac{1}{2\Delta z^2} (-\kappa_{i+1} - 2\kappa_i - \kappa_{i-1}) \right) \\ &+ C_{i-1} \left(\frac{1}{2\Delta z^2} (-\kappa_{i-1} - \kappa_i) \right) \end{aligned} \tag{10}$$

Discretizing in time

Next we discretize

$$\frac{\partial C}{\partial t} + w_s \frac{\partial C}{\partial z} = \frac{\partial}{\partial z} \left(\kappa_t \frac{\partial C}{\partial z} \right) + \gamma C$$

in time. We use an implicit scheme,

$$\frac{C_i^{n+1} - C_i^n}{\Delta t} + w_s \frac{C_i^{n+1} - C_{i-1}^{n+1}}{\Delta z} = \left(\kappa_t \frac{\partial C}{\partial z} \right)^{n+1} + \gamma C_i^{n+1}$$

We plug in our expression for the Laplacian discretized in space (Eq. 10), evaluated at time $n+1$,

$$\begin{aligned} \frac{C_i^{n+1} - C_i^n}{\Delta t} + w_s \frac{C_i^{n+1} - C_{i-1}^{n+1}}{\Delta z} &= C_{i+1}^{n+1} \left(\frac{1}{2\Delta z^2} (\kappa_{i+1}^n + \kappa_i^n) \right) \\ &+ C_i^{n+1} \left(\frac{1}{2\Delta z^2} (-\kappa_{i+1}^n - 2\kappa_i^n - \kappa_{i-1}^n) \right) \\ &+ C_{i-1}^{n+1} \left(\frac{1}{2\Delta z^2} (-\kappa_{i-1}^n - \kappa_i^n) \right) \\ &+ \gamma C_i^{n+1} \end{aligned}$$

$$\begin{aligned}
C_i^{n+1} - C_i^n + \Delta t w_s \frac{C_i^{n+1} - C_{i-1}^{n+1}}{\Delta z} &= C_{i+1}^{n+1} \left(\frac{\Delta t}{2\Delta z^2} (\kappa_{i+1}^n + \kappa_i^n) \right) \\
&+ C_i^{n+1} \left(\frac{\Delta t}{2\Delta z^2} (-\kappa_{i+1}^n - 2\kappa_i^n - \kappa_{i-1}^n) \right) \\
&+ C_{i-1}^{n+1} \left(\frac{\Delta t}{2\Delta z^2} (-\kappa_{i-1}^n - \kappa_i^n) \right) \\
&+ \Delta t \gamma C_i^{n+1}
\end{aligned}$$

This all simplifies out to,

$$\begin{aligned}
C_i^n &= C_{i+1}^{n+1} \left(-\frac{\Delta t}{2\Delta z^2} (\kappa_i^n + \kappa_{i+1}^n) \right) \\
&+ C_i^{n+1} \left(1 + \gamma_i^n + \frac{\Delta t w_s}{\Delta z} + \frac{\Delta t}{2\Delta z^2} (\kappa_{i+1}^n + 2\kappa_i^n + \kappa_{i-1}^n) - \Delta t \gamma \right) \\
&+ C_{i-1}^{n+1} \left(-\frac{\Delta t w_s}{\Delta z} - \frac{\Delta t}{2\Delta z^2} (\kappa_{i-1}^n + \kappa_i^n) \right)
\end{aligned}$$

In code, to solve this tridiagonal system, we initialize three vectors A, B, and C to represent the diagonals of the matrix.

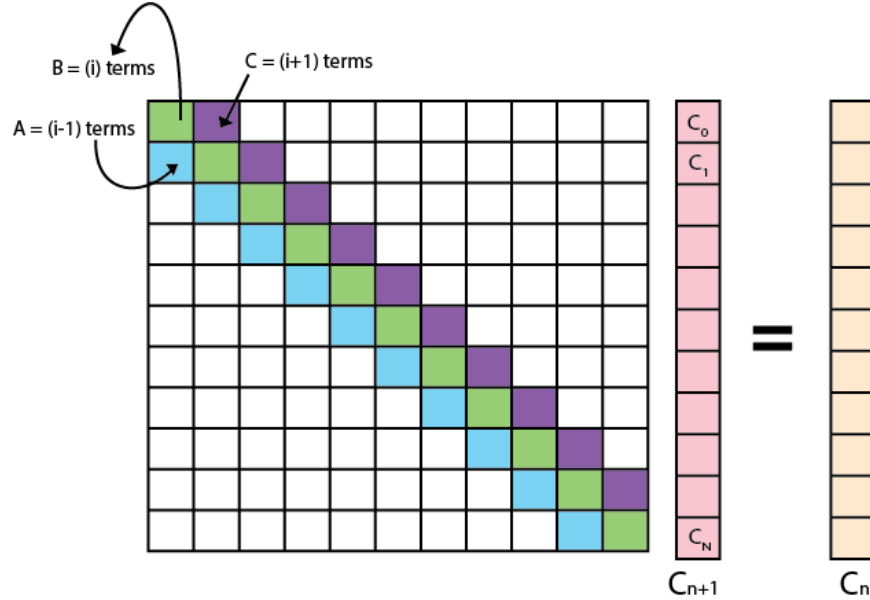


Figure 2: Tridiagonal matrix for implicit scheme

```

# Initialize empty arrays for tridiagonals of algae matox
aA, bA, cA, dA = lib.initialize_abcd(N)

beta = dt / dz**2
# Set up vectors (a --> i+1 ; b --> main diagonal, i ; c --> i-1)
ws = diatoms.ws
dtdz = dt/dz
gamma = diatoms.get_loss_and_growth(I_in = 350, current_concentration = Ap) # Growth +

aA[1:top] = - beta/2 * (Kzp[0:top-1] + Kzp[1:top])
bA[1:top] = 1 + ws*dtdz + beta/2 * (Kzp[2:top+1] + \
    2*Kzp[1:top] + Kzp[0:top-1]) - gamma[1:top]*dt
cA[1:top] = -ws*dtdz - beta/2 * (Kzp[1:top] + Kzp[2:top+1])
dA = Ap # RHS, C_n

```

We now need to consider the boundary conditions. Because we are considering a scalar value, we want zero flux at the bed and at the top.

$$\frac{C_N - C_{N+1}}{\Delta z} = 0$$

$$C_N = C_{N+1}$$

and by the same logic,

$$C_0 = C_{-1}$$

Zero-flux at top of water column

Setting these terms equal in the equation,

$$\begin{aligned}
C_N^n &= C_N^{n+1} \left(-\frac{\Delta t}{2\Delta z^2} (\kappa_i^n + \kappa_{i+1}^{n+1}) \right) \\
&+ C_N^{n+1} \left(1 + \gamma_i^n + \frac{\Delta t w_s}{\Delta z} + \frac{\Delta t}{2\Delta z^2} (\kappa_{i+1}^n + 2\kappa_i^n + \kappa_{i-1}^n) - \Delta t \gamma \right) \\
&+ C_{N-1}^{n+1} \left(-\frac{\Delta t w_s}{\Delta z} - \frac{\Delta t}{2\Delta z^2} (\kappa_{i-1}^n + \kappa_i^n) \right)
\end{aligned}$$

we see that several terms cancel,

$$\begin{aligned}
C_N^n &= C_N^{n+1} \left(1 + \gamma_i^n + \frac{\Delta t w_s}{\Delta z} + \frac{\Delta t}{2\Delta z^2} (\kappa_i^n + \kappa_{i-1}^n) - \Delta t \gamma \right) \\
&+ C_{N-1}^{n+1} \left(-\frac{\Delta t w_s}{\Delta z} - \frac{\Delta t}{2\Delta z^2} (\kappa_{i-1}^n + \kappa_i^n) \right)
\end{aligned}$$

In code, we impose this like so,

```

aA[top] = -ws*dtdz - beta/2 * (Kzp[top] + Kzp[top-1])
bA[top] = 1 + ws*dtdz + beta/2 * (Kzp[top] + Kzp[top-1]) - gamma[top]*dt
dA[top] = Ap[top]

```

Zero-flux at bed of water column

Now setting terms equal at $i = 0$, the bed,

$$\begin{aligned}
C_0^n &= C_1^{n+1} \left(-\frac{\Delta t}{2\Delta z^2} (\kappa_i^n + \kappa_{i+1}^{n+1}) \right) \\
&+ C_0^{n+1} \left(1 + \gamma_i^n + \frac{\Delta t w_s}{\Delta z} + \frac{\Delta t}{2\Delta z^2} (\kappa_{i+1}^n + 2\kappa_i^n + \kappa_{i-1}^n) - \Delta t \gamma \right) \\
&+ C_0^{n+1} \left(-\frac{\Delta t w_s}{\Delta z} - \frac{\Delta t}{2\Delta z^2} (\kappa_{i-1}^n + \kappa_i^n) \right)
\end{aligned}$$

$$C_0^n = C_1^{n+1} \left(-\frac{\Delta t}{2\Delta z^2} (\kappa_i^n + \kappa_{i+1}^{n+1}) \right) + C_0^{n+1} \left(1 + \gamma_i^n + \frac{\Delta t}{2\Delta z^2} (\kappa_{i+1}^n + \kappa_i^n) - \Delta t \gamma \right)$$

In code, we impose this like so,

```

bA[0] = 1 + beta/2 * (Kzp[1] + Kzp[0]) - gamma[0]*dt # + ws*dtdz
cA[0] = -beta/2 * (Kzp[1] + Kzp[0])
dA[0] = Ap[0]

```

Phytoplankton Growth

Now that we have an equation for advection-diffusion of phytoplankton, we can dig into the growth and loss terms. Here, I'll follow the conventions set forth in Huisman's paper.

They discretize the light dependence of phytoplankton growth using a Monod function,

$$p(I) = p_{max} \frac{I}{H + I}$$

H is the half-saturation constant of light limited growth and p_{max} is the maximum specific production rate for the species. Physically, this means that growth at any given point in the water column is the max-possible growth scaled by a proportion of the amount of available light I and the species' sensitivity to light availability H .

The light available at any point in the water column is a function of the amount of algae above it (self-shading) and background turbidity. They express this using Lambert-Beer's law applied to nonuniform population density distributions:

$$I(z, t) = I_{in} \exp \left(- \int_0^z \left(\sum_{j=1}^n k_j \omega_j(\sigma, t) \right) d\sigma - K_{bg} z \right) \quad (11)$$

where I_{in} is incident light intensity at the top of the water column, k_j is the specific light attenuation coefficient of the phytoplankton species j ; K_{bg} is the background turbidity (eg, not phytoplankton) and σ is an integration variable.

In code, we calculate that with the following function:

```
def self_shading(ListofAlgae, I_in, turbidity):
    ''' Use Lambert-Beer's Law to calculate light intensity at each depth '''
    corrected_z = (-1) * z # We want water depth [cm] to be zero
    # at the top, 20 at the bottom (positive numbers)
    kxC = np.zeros(N)
    for species in ListofAlgae:
        kxC += species.k * species.c

    I = np.zeros(N)
    vector = kxC - (turbidity*corrected_z)
    for i in (range(N)):
        I[i] = np.sum(vector[i:-1]) * z[i]

    I[I<0] = 1e-10
    I = I_in * np.exp(-I)
```

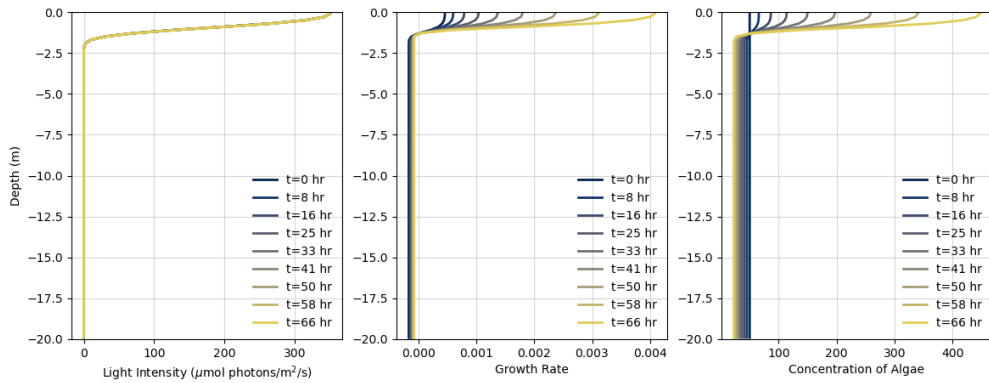


Figure 3: Algae Concentration with constant light

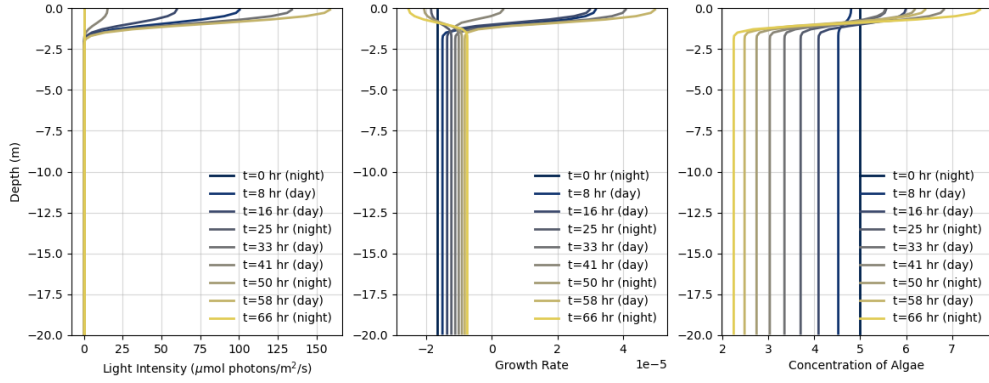


Figure 4: Algae Concentration with diurnal light

We can write the population for phytoplankton C as

$$\frac{\partial C}{\partial t} + w_s \frac{\partial C}{\partial z} = \kappa \frac{\partial^2 C}{\partial z^2} + \Gamma(z)C \quad (12)$$

where $\Gamma(z)$ is the source + sink term (e.g., mortality + growth).

$$\Gamma(z) = p_i(I) - L_i$$

and $I(z, t)$ is the light intensity at depth z and time t . Thus, we expect Γ to vary as a function of depth and time.

Monod equation for phytotop

Huisman's equation:

$$\frac{\partial \omega_i}{\partial t} = p_i(I)\omega_i - L_i\omega_i + v_i \frac{\partial \omega_i}{\partial z} + D \frac{\partial^2 \omega_i}{\partial z^2} \quad (13)$$

The boundary condition for phytoplankton at the top and bottom of the water column is zero-flux:

$$v_i \omega_i(z, t) + D \frac{\partial \omega_i}{\partial z} = 0 \big|_{z=0}$$

Problem 3

Write the equation of motion for a velocity fluctuation.

We start by recalling the equation of motion for momentum, after performing Reynold's decomposition and splitting components into their mean and fluctuating parts...

$$\partial_t(\bar{u}_j + u'_j) + (\bar{u}_i + u'_i)\partial_i(\bar{u}_j + u'_j) = -\frac{1}{\rho}\partial_j(\bar{P} + P') + \nu\partial_k\partial_k(\bar{u}_j + u'_j)$$

$$\partial_t\bar{u}_j + \partial_t u'_j + \bar{u}_i\partial_i\bar{u}_j + \bar{u}_i\partial_i u'_j + u'_i\partial_i\bar{u}_j + u'_i\partial_i u'_j = -\frac{1}{\rho}\partial_j\bar{P} - \frac{1}{\rho}\partial_j P' + \nu\partial_k\partial_k\bar{u}_j + \nu\partial_k\partial_k u'_j \quad (14)$$

Now we can subtract our RANS equation for momentum from this equation to isolate the turbulent components (u'). Recall our RANS equation for momentum reads

$$\partial_t\bar{u}_j + \bar{u}_i\partial_i\bar{u}_j = -\frac{1}{\rho}\partial_j\bar{P} + \nu\partial_k\partial_k\bar{u}_j - \partial_i\overline{u'_j u'_i}. \quad (15)$$

Subtracting Equation (2) from Equation (1) yields

$$\partial_t u'_j + \bar{u}_i\partial_i u'_j + u'_i\partial_i\bar{u}_j + u'_i\partial_i u'_j = -\frac{1}{\rho}\partial_j P' + \nu\partial_k\partial_k u'_j + \partial_i\overline{u'_j u'_i}. \quad (16)$$

$$\partial_t u'_j + \bar{u}_i\partial_i u'_j = -\frac{1}{\rho}\partial_j P' + \nu\partial_k\partial_k u'_j + \partial_i\overline{u'_j u'_i} - u'_i\partial_i\bar{u}_j - u'_i\partial_i u'_j. \quad (17)$$

The TKE Equation can be expressed as

$$\frac{\partial k}{\partial t} + \bar{U}_j \quad (18)$$

The density profile is calculated according to

$$\rho = \rho_0(1 - \alpha(C - T_0)) \quad (19)$$

Turbulent kinetic energy equation

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z} \left(k_q \frac{\partial q^2}{\partial z} \right) + 2P + 2B - 2\epsilon$$

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z} \left(k_q \frac{\partial q^2}{\partial z} \right) + \nu_t \frac{\partial u^2}{\partial z} + 2 \frac{g}{\rho_0} \kappa_p \frac{\partial p}{\partial z} - 2 \frac{(q^2)^{3/2}}{B_1 l}$$

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z} \left(k_q \frac{\partial q^2}{\partial z} \right) + \underbrace{\nu_t \left(\frac{\partial u}{\partial z} \right)^2}_{\text{shear production}} + \underbrace{2 \frac{g}{\rho_0} \kappa_p \frac{\partial p}{\partial z}}_{\text{bouyancy production}} - \underbrace{2 \frac{(q^2)^{3/2}}{B_1 l}}_{\text{dissipation}}$$

We will think of barotropic pressure is $f(t)$ and barolinic is $f(t, z)$. We will write this gradient $\frac{\partial p}{\partial x}$ as P_x , the pressure gradient in the x-direction. We can now think about discretizing this equation,

$$\frac{\partial q^2}{\partial t} = \underbrace{\frac{\partial}{\partial z} \left(k_q \frac{\partial q^2}{\partial z} \right)}_{n+1} + \underbrace{\nu_t \left(\frac{\partial u}{\partial z} \right)^2}_n + \underbrace{2 \frac{g}{\rho_0} \kappa_p \frac{\partial p}{\partial z}}_n - \underbrace{2 \frac{(q^2)^{3/2}}{B_1 l}}_{\text{mixed}}$$

Start to discretize...($Q = q^2$)

$$\frac{Q_i^{n+1} - Q_i^n}{\Delta t} = \frac{1}{\Delta z} \left(k_q \frac{Q_i^{n+1} - Q_{i-1}^{n+1}}{\Delta z} \right) + (\nu_t)_i^n \left(\left(\frac{\partial u}{\partial z} \right)_i^n \right)^2 + 2 \frac{g}{\rho_0} (\kappa_p)_i^n \frac{\partial p}{\partial z} - 2 \frac{(q^2)^{3/2}}{B_1 l}$$

We write the dissipation as a mixed discretization,

$$-2 \left(\frac{\sqrt{Q}}{B_1 l} \right)_i^n \cdot Q_i^{n+1}$$

where it's partially linearized and partially explicit.

We start with the continuity equation,

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i} (\rho U_i) = 0 \quad (20)$$

where flow is considered to be divergent-free, or solenoidal.

Meanwhile, the Reynolds-Averaged Navier Stokes reads:

$$\frac{\partial \bar{U}}{\partial t} + \bar{U}_j \frac{\partial \bar{U}_k}{\partial x_j} + \rho \epsilon_{jkl} f_k U_l = \frac{\partial}{\partial x_k} \overline{u_k u_j} - \frac{\partial \bar{P}}{\partial x_j} - g \hat{e}_3 \quad (21)$$

The Boussinesq approximation is invoked to represent that density fluctuations affect buoyancy but no other transport/state terms. These fluctuations are written $\beta\theta$, where β is the coefficient of thermal expansion.

Mellor and Herring (1973) uses the following equation:

$$\left\langle \frac{p}{\rho} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right\rangle = -\frac{q}{3l_1} (\overline{u_i u_j} - \frac{\delta_{ij}}{3} q^2) + C_1 q^2 \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right) \quad (22)$$

We are establishing equations for the following: $\overline{u_k u_i u_j}$, $\overline{p u_j}$, $\overline{u_i u_j \theta}$, and $\overline{p \theta}$

Math Note The velocity gradient $\frac{\partial U_j}{\partial x_i}$ can be decomposed into symmetric and antisymmetric parts,

$$\frac{\partial U_j}{\partial x_i} = S_{ij} + \Omega_{ij}$$

where the symmetric rate-of-strain tensor is

$$S_{ij} = \frac{1}{2} \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right)$$

and the asymmetric part is,

$$\Omega_{ij} = \frac{1}{2} \left(\frac{\partial U_i}{\partial x_j} - \frac{\partial U_j}{\partial x_i} \right)$$

Parameterization

$$\langle u_k u_i u_j \rangle = \frac{3}{5} l q S_q \left(\frac{\partial \langle u_i u_j \rangle}{\partial x_k} + \frac{\partial \langle u_i u_k \rangle}{\partial x_j} + \frac{\partial \langle u_j u_k \rangle}{\partial x_i} \right) \quad (23)$$

In this equation S_q is a dimensionless number.

Sm :

$$\frac{A_1}{A_2} \frac{B_1(\gamma - C_1) - (B_1(\gamma_1 - C_1) + 6(A_1 + 3A_2))R_f}{B_1\gamma_1 - (B_1(\gamma_1 + \gamma_2) - 3A_1)R_f} \quad (24)$$

where

$$\gamma_1 = \frac{1}{3} - \frac{2A_1}{B_1}$$

and

$$\gamma_2 = \frac{B_2}{B_1} - \frac{6A_1}{B_1}$$

Sh:

$$S_h = \frac{A_2(1 - \frac{6A_1}{B_1})}{1 - (3A_2B_2G_h + 18A_1A_2G_h)} = \frac{A_2(1 - \frac{6A_1}{B_1})}{(1 - 3A_2G_h(B_2 + 6A_1))}$$

$$S_m = \frac{A_1(1 - 3C_1 - \frac{6A_1}{B_1}) + S_HG_h(18A_1 + 9A_1A_2)}{(1 - 9A_1A_2G_h)}$$

Richardson's number:

$$G_H = -\frac{L^2}{Q^2} \frac{g}{\rho_0} \frac{\partial}{\partial z}$$

Chosen parameters

$$C_1 = \gamma_1 - \frac{1}{3A_1} B_1^{1/3} \approx 0.08$$

$$B_2 = \frac{B_1^{1/3}}{P_{rt}} \frac{u_T^2 \langle \theta^2 \rangle}{H^2} \approx$$

$$B_1 = 16.6$$

writing equations based on code

$$bX = bX - \frac{aXcX}{bX}$$

$$dX = dX - \frac{aXdX}{bX}$$

$$u_* = U_{bed} \sqrt{C_d}$$

$$G_h = - \left(\frac{N_{BV} * L}{Q + Small} \right)^2$$

$$diss = 2 \frac{dt * \sqrt{Q^2}}{B_1 L_p}$$